

Translated

From: e-maxx.ru

Finding the equation of a line for a segment

The task is: given the coordinates of the ends of a segment, construct a line passing through it.

We assume that the segment is non-degenerate, i.e. has a length greater than zero (otherwise, of course, infinitely many different lines pass through it).

Two-dimensional case

Let the given segment be PQ i.e. the known coordinates of its ends P_x, P_y, Q_x, Q_y .

It is necessary to construct **the equation of a line in the plane** passing through this segment, i.e. find the coefficients A, B, C in the equation of a line:

$$Ax + By + C = 0.$$

Note that for the required triples (A, B, C) there are **infinitely many** solutions which describe the given segment: you can multiply all three coefficients by an arbitrary non-zero number and get the same straight line. Therefore, our task is to find one of these triples.

It is easy to verify (by substitution of these expressions and the coordinates of the points P and Q into the equation of a straight line) that the following set of coefficients fits:

$$\begin{aligned} A &= P_y - Q_y, \\ B &= Q_x - P_x, \\ C &= -A P_x - B P_y. \end{aligned}$$

Integer case

An important advantage of this method of constructing a straight line is that if the coordinates of the ends were integer, then the coefficients obtained will also be **integer**. In some cases, this allows one to perform geometric operations without resorting to real numbers at all.

However, there is a small drawback: for the same straight line different triples of coefficients can be obtained. To avoid this, but do not go away from the integer coefficients, you can apply the following technique, often called **rationing**. Find the **greatest common divisor** of numbers $|A|, |B|, |C|$, we divide all three coefficients by it, and then we make the normalization of the sign: if $A < 0$ or $A = 0, B < 0$ then multiply all three coefficients by -1 . As a result, we will come to the conclusion that for identical straight lines, identical triples of coefficients will be obtained, which makes it easy to check straight lines for equality.